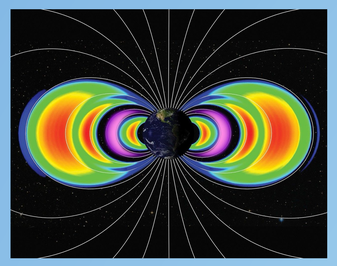
Analyzing the Fine Structure of Chorus Waves in Response to the Kennel-Petschek Limit

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I. Introduction

The Van Allen Probes mission has been orbiting since August 30, 2012, with a goal of exploring the Earth's Radiation Belts, two toroidal regions containing highly energetic protons and electrons.



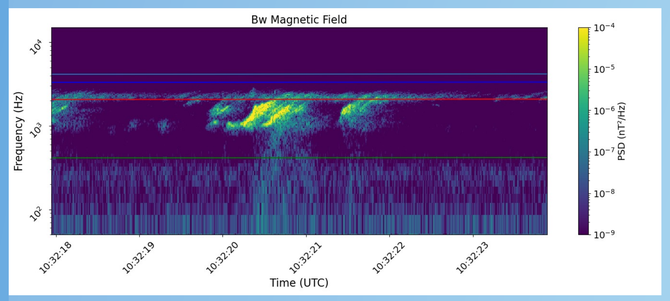
Kennel-Petschek Process [2]

Big particle injections → Electron Fluxes exceed the K-P Limit threshold → Intense chorus wave generation, Particles diffused → Balance reached at K-P Limit

We focus on understanding how **electron fluxes** and **pitch angle distributions** in the outer radiation belt (L =4.0 – 4.5) change in response to chorus wave generation .

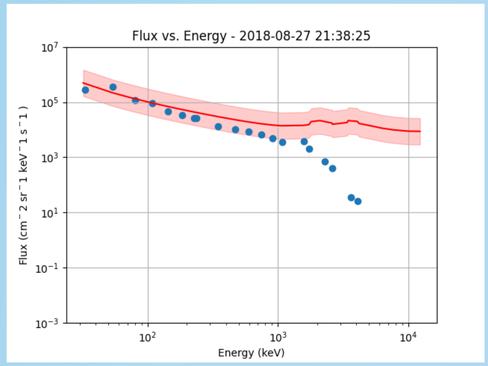
II. Data and Methods

We use survey and burst-mode data from EMFISIS (Electric and Magnetic Field Instrument Suite and Integrated Science).



- Burst-mode data allows us to see the fine-scale details in chorus waves. We calculated the electron cyclotron frequency from EMFISIS magnetic field data as :

$$f_{ce} = \frac{eB}{2\pi m_e}$$



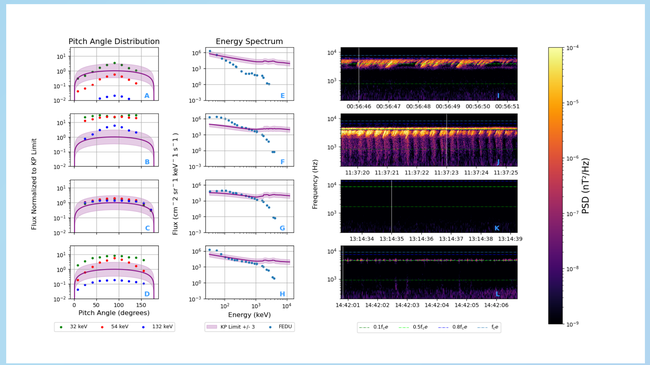
- For a given moment in time, we can analyze the 90-degree differential electron flux compared to the KP-Limit. In conjunction with further context, such as the pitch angle distribution and the wave spectrograms at that time, we gain insight into the radiation belt dynamics.

IV. Conclusions

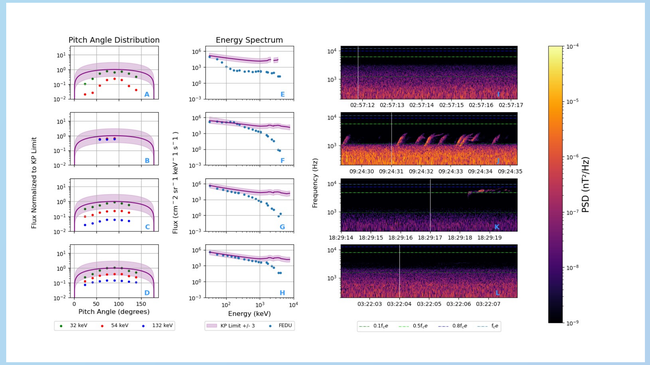
- For all selected geomagnetic storms, **chorus wave observations stopped** while electron fluxes returned to below the KP-Limit
- Consistent with prior studies [1], >1 MeV electrons remain **unaffected** by the KP-Limit.
- **32 keV** electron fluxes being at or above the KP-limit is the factor which most often coincides with chorus wave observations.

III. Results

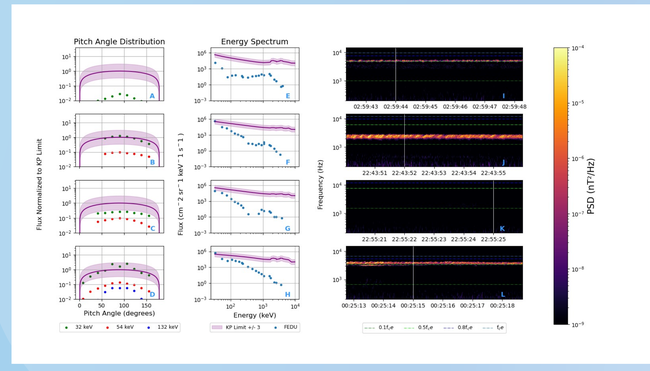
We have selected three geomagnetic storms during the Van Allen Probes era to examine different flux behaviors – the first storm which greatly exceeds the KP-Limit, one which very closely adheres to the KP-Limit, and finally one which remains largely below the KP-Limit.



- Electron fluxes up to **1 MeV** elevated near the KP-Limit
- 2 particle injection events: chorus waves visible whilst **fluxes spike above 3 x KP-Limit**
- [I] : Chorus waves cover a **wide frequency** range, with some bursts extending below 0.5 f_{ce}
- [J] : Waves in **two distinct bands–** one continuous above 0.5 f_{ce} , evenly-spaced lower band waves with decreasing intensity.
- Anisotropic pitch angle distribution [A,B,D] during chorus wave observations – peaked at 90° for injected electron populations, **butterfly** in <54 keV populations [B] **while chorus waves** are being generated



- Electron fluxes never exceed 3xKP-Limit.
- Lower band chorus waves visible while **~100-132 keV** electrons are above the KP-Limit
  - **Non-uniform** spacing, lower PSD in waves
- Upper band chorus waves visible while **32 keV electrons** are above the KP-Limit
  - Waves **rising** in frequency from just below 0.5 f_{ce} to bordering 0.8 f_{ce}
- In periods where fluxes drop to **at or below the KP-Limit**, no waves are detected.
- [A] – 32 keV electrons show **butterfly** PAD, 54 keV PAD is **flat-topped**



- [I] – Bursty waves occupy a narrow frequency range, centered at 0.5 f_{ce}
- Chorus waves visible **whilst only 32 keV** electrons are **above** the KP-limit
  - [J] waves **uniform** in shape, frequency, and intensity occupying a narrow portion of the **lower band**
  - [L] **Upper band** waves occupying a wider portion of the band, weaking in intensity.
- Electrons >75 keV do not reach the shaded ±3 KP-Limit region for the duration of the storm.
- **Butterfly** pitch angle distribution for 32 keV electrons simultaneous to **upper band chorus wave** observations, whilst only 32 keV electrons are above the KP-Limit

References:

1. L. Olifer et al. (2022), A Natural Limit to the Spectral Hardness of Worst Case Electron Radiation in the Terrestrial Van Allen Belt
2. C.F Kennel, H.E Petschek (1966), A limit on stably trapped particle fluxes